

Kazakh British Technical University

Databases

Laboratory Work №2

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Task 1: Normalization

1. Normal Form Violation The original `Booking_receipt` table violates the **First Normal Form (1NF)**. **Why:** The `seat_numbers` column contains multiple values in one cell (like "12A, 12B"). According to 1NF rules, every cell must have only one single (atomic) value.

2. Step-by-Step Normalization *(Primary keys are underlined, Foreign keys are marked as FK)*

Step 1: First Normal Form (1NF) To fix the 1NF violation, I separated the seat numbers. Now, each seat has its own row. Because of this, I had to create a composite primary key using two columns.

```
Booking_receipt_1NF(booking_id, seat_number, passenger_full_name,  
passenger_passport_number, flight_number, departure_airport_name, departure_city,  
arrival_airport_name, arrival_city, airline_name, ticket_price)
```

Step 2: Second Normal Form (2NF) Next, I checked for partial dependencies. In my 1NF table, details like passenger name or flight number only depend on the `booking_id`, not on the `seat_number`. So, I split the data into two tables:

```
Booking_2NF(booking_id, passenger_full_name, passenger_passport_number, flight_number,  
departure_airport_name, departure_city, arrival_airport_name, arrival_city, airline_name,  
ticket_price)
```

```
Booking_Seat_2NF(booking_id (FK), seat_number)
```

Step 3: Third Normal Form (3NF) Finally, I removed transitive dependencies from the `Booking_2NF` table. For example, passenger name depends on the passport number, not directly on the booking. I created separate reference tables to avoid data duplication:

```
Passenger_3NF(passport_number, passenger_full_name)
```

```
Airport_3NF(airport_name, city)
```

```
Flight_3NF(flight_number, departure_airport_name (FK), arrival_airport_name (FK),  
airline_name)
```

```
Booking_3NF(booking_id, passport_number (FK), flight_number (FK), ticket_price)
```

```
Booking_Seat_3NF(booking_id (FK), seat_number)
```

3. Functional Dependencies I used these logical rules to split my tables:

`booking_id` → `passenger_passport_number`, `flight_number`, `ticket_price`

`passenger_passport_number` → `passenger_full_name`

`flight_number` → `departure_airport_name`, `arrival_airport_name`, `airline_name`

departure_airport_name → departure_city

arrival_airport_name → arrival_city

4. Checking the Flights Table (from DDL) The `Flights` table from the DDL task is already in 3NF.

Primary Key: `flight_id`

Why it is in 3NF: There are no transitive dependencies. Instead of storing full names of airports or airlines (which would duplicate data), the table only stores their IDs (`departing_airport_id`, `airline_id`). All non-key columns depend exactly on the `flight_id` and nothing else.

Task 5: SQL Implementation of 3NF Tables and Data Insertion

1. Creating 3NF Tables

```
CREATE TABLE Passenger_3NF (  
    passport_number VARCHAR(20) PRIMARY KEY,  
    passenger_full_name VARCHAR(100) NOT NULL  
);
```

```
CREATE TABLE Airport_3NF (  
    airport_name VARCHAR(100) PRIMARY KEY,  
    city VARCHAR(50) NOT NULL  
);
```

```
CREATE TABLE Flight_3NF (  
    flight_number VARCHAR(20) PRIMARY KEY,  
    departure_airport_name VARCHAR(100) REFERENCES Airport_3NF(airport_name),  
    arrival_airport_name VARCHAR(100) REFERENCES Airport_3NF(airport_name),  
    airline_name VARCHAR(50) NOT NULL  
);
```

```
CREATE TABLE Booking_3NF (  
    booking_id INT PRIMARY KEY,  
    passport_number VARCHAR(20) REFERENCES Passenger_3NF(passport_number),
```

```
flight_number VARCHAR(20) REFERENCES Flight_3NF(flight_number),  
ticket_price DECIMAL(7,2) NOT NULL  
);
```

```
CREATE TABLE Booking_Seat_3NF (  
    booking_id INT REFERENCES Booking_3NF(booking_id),  
    seat_number VARCHAR(10),  
    PRIMARY KEY (booking_id, seat_number)  
);
```

2. Inserting at least 5 rows of sample data

```
INSERT INTO Passenger_3NF (passport_number, passenger_full_name) VALUES  
( 'N123456', 'John Doe'),  
( 'N654321', 'Jane Smith'),  
( 'K987654', 'Alice Johnson'),  
( 'L345678', 'Bob Brown'),  
( 'M555555', 'Charlie Davis');
```

```
INSERT INTO Airport_3NF (airport_name, city) VALUES  
( 'JFK International', 'New York'),  
( 'Heathrow', 'London'),  
( 'Almaty International', 'Almaty'),  
( 'Charles de Gaulle', 'Paris');
```

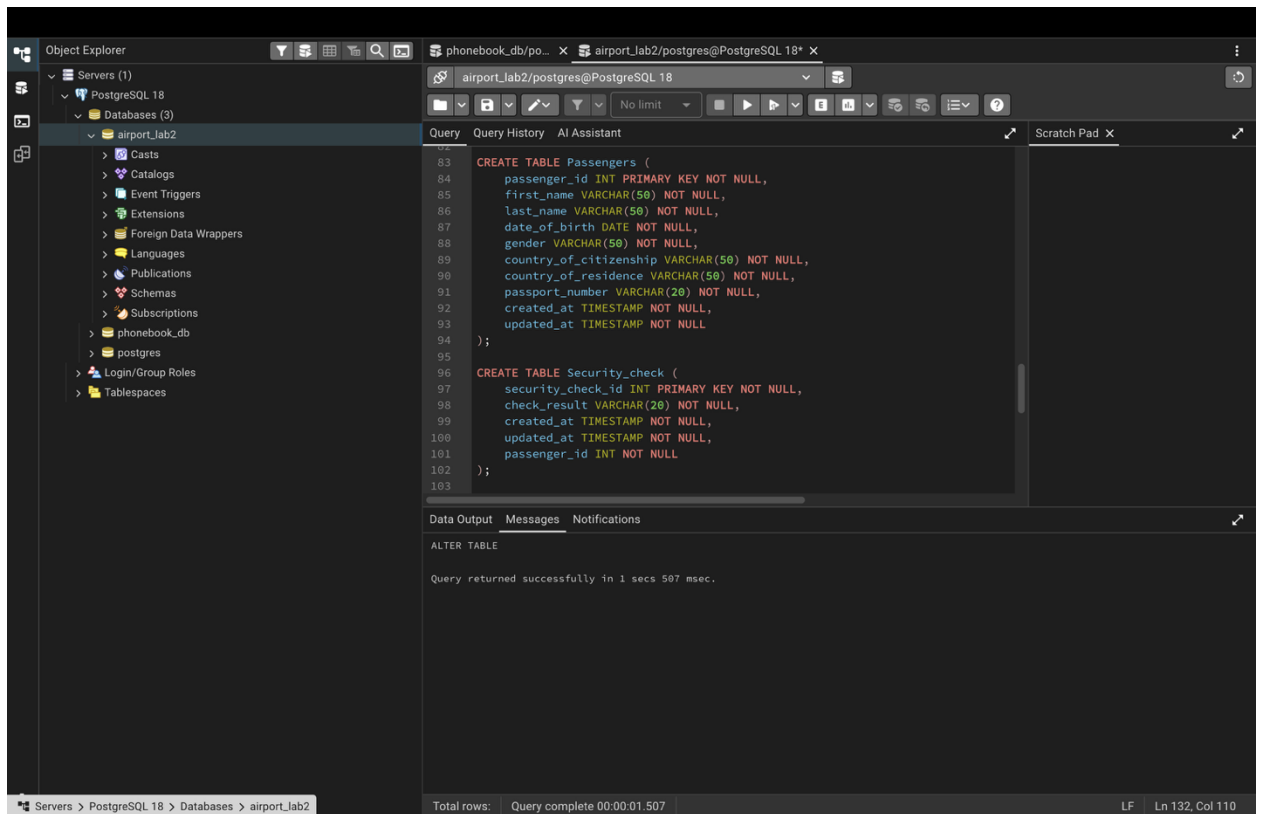
```
INSERT INTO Flight_3NF (flight_number, departure_airport_name, arrival_airport_name,  
airline_name) VALUES  
( 'BA101', 'JFK International', 'Heathrow', 'British Airways'),  
( 'KC909', 'Almaty International', 'Heathrow', 'Air Astana'),  
( 'AF202', 'Charles de Gaulle', 'JFK International', 'Air France');
```

```
INSERT INTO Booking_3NF (booking_id, passport_number, flight_number, ticket_price)  
VALUES
```

(1, 'N123456', 'BA101', 500.00),
 (2, 'N654321', 'KC909', 450.00),
 (3, 'K987654', 'AF202', 600.00),
 (4, 'L345678', 'BA101', 500.00),
 (5, 'M555555', 'KC909', 450.00);

INSERT INTO Booking_Seat_3NF (booking_id, seat_number) VALUES

(1, '12A'),
 (1, '12B'),
 (2, '14C'),
 (3, '1A'),
 (4, '15F'),
 (5, '22B');



Object Explorer

- Servers (1)
 - PostgreSQL 18
 - Databases (3)
 - airport_lab2
 - Casts
 - Catalogs
 - Event Triggers
 - Extensions
 - Foreign Data Wrappers
 - Languages
 - Publications
 - Schemas
 - Subscriptions
 - phonebook_db
 - postgres
 - Login/Group Roles
 - Tablespaces

Query

```
112
113
114 -- Passengers relationships
115 ALTER TABLE Security_check ADD CONSTRAINT fk_sec_pass FOREIGN KEY (passenger_id) RE
116 ALTER TABLE Baggage_check ADD CONSTRAINT fk_bagcheck_pass FOREIGN KEY (passeng
117
118 -- Booking relationships
119 ALTER TABLE Baggage_check ADD CONSTRAINT fk_bagcheck_book FOREIGN KEY (booking
120 ALTER TABLE Baggage ADD CONSTRAINT fk_baggage_book FOREIGN KEY (booking_id) RE
121 ALTER TABLE Boarding_pass ADD CONSTRAINT fk_board_book FOREIGN KEY (booking_id
122 ALTER TABLE Booking_flight ADD CONSTRAINT fk_bookflight_book FOREIGN KEY (book
123
124 -- Flights relationships
125 ALTER TABLE Booking_flight ADD CONSTRAINT fk_bookflight_flight FOREIGN KEY (fl
126
127 -- Airport relationships
128 ALTER TABLE Flights ADD CONSTRAINT fk_flight_dep_airport FOREIGN KEY (departir
129 ALTER TABLE Flights ADD CONSTRAINT fk_flight_arr_airport FOREIGN KEY (arriving
130
131 -- Airline relationships
132 ALTER TABLE Flights ADD CONSTRAINT fk_flight_airline FOREIGN KEY (airline_id)
```

Data Output Messages Notifications

ALTER TABLE

Query returned successfully in 1 secs 507 msec.

Total rows: Query complete 00:00:01.507 LF Ln 132, Col 110

Object Explorer

- Servers (1)
 - PostgreSQL 18
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 - airport_lab2
 - Casts
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 - Extensions
 - Foreign Data Wrappers
 - Languages
 - Publications
 - Schemas (1)
 - public
 - Aggregates
 - Collations
 - Domains
 - FTS Configurations
 - FTS Dictionaries
 - FTS Parsers
 - FTS Templates
 - Foreign Tables
 - Functions
 - Materialized Views
 - Operators
 - Procedures
 - Sequences
 - Tables (15)
 - airline
 - airport
 - airport_3nf
 - baggage
 - baggage_check
 - boarding_pass
 - booking
 - booking_3nf
 - booking_flight
 - booking_seat_3nf
 - flight_3nf

Query

```
38 ('M3333333', 'Charlie Davis');
39
40 INSERT INTO Airport_3NF (airport_name, city) VALUES
41 ('JFK International', 'New York'),
42 ('Heathrow', 'London'),
43 ('Almaty International', 'Almaty'),
44 ('Charles de Gaulle', 'Paris');
45
46 INSERT INTO Flight_3NF (flight_number, departure_airport_name, arrival_airport
47 ('BA101', 'JFK International', 'Heathrow', 'British Airways'),
48 ('KC909', 'Almaty International', 'Heathrow', 'Air Astana'),
49 ('AF202', 'Charles de Gaulle', 'JFK International', 'Air France');
50
51 INSERT INTO Booking_3NF (booking_id, passport_number, flight_number, ticket_pr
52 (1, 'N123456', 'BA101', 500.00),
53 (2, 'N654321', 'KC909', 450.00),
54 (3, 'K987654', 'AF202', 600.00),
55 (4, 'L345678', 'BA101', 500.00),
56 (5, 'M555555', 'KC909', 450.00);
57
58 INSERT INTO Booking_Seat_3NF (booking_id, seat_number) VALUES
59 (1, '12A'),
```

Data Output Messages Notifications

INSERT 0 6

Query returned successfully in 1 secs 745 msec.

Total rows: Query complete 00:00:02.430 LF Ln 64, Col 12